Econ 812 Recitation 11 Notes

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1 A Two-Sector Model with Investment Shocks June 22

The representative household has preferences of the final good and labor according to

$$E_0 \sum_{t=0}^{\infty} \beta^t (1/2) C_t^{1/2}$$

The household rents capital K_t at rate r and has a unit of time that it supplies inelastically as labor N_t at wage w_t to firms (final good and investment good). The household period t budget constraint is

$$C_t + P_t I_t = w_t + r_t K_t + \Pi_t$$

where C_t is consumption of the consumption good, It is the investment good, P_t is the price of the investment good (in units of final good), and Π_t is any profit from the firms. The law of motion for capital is

$$K_{t+1} = I_t$$

The consumption good firm seeks to maximize profit

$$\pi_{yt} = K_t^{\alpha} N_{yt}^{1-\alpha} - r_t K_t - w_t N_{yt}$$

by choice of K_t, N_{yt} taking r_t, w_t as given with $0 < \alpha < 1$. The investment good firm seeks to maximize profit

$$\pi_{it} = P_t A_t N_{it} - w_t N_{it}$$

by choice of N_{it} taking P_t,w_t as given as well as the law of motion for A_t:

$$\ln A_{t+1} = \rho \ln A_t + e_t$$

where $|\rho| < 1$ and e_t is iid mean zero with bounded support.

1.1 Part A

Define a competitive equilibrium for this model. [5 pts]

A symmetric competitive equilibrium is a set of prices $P = \{w_t, r_t, P_t\}$, a plan for the firms $F = \{k_t, n_{yt}, n_{it}, \pi_{yt}, \pi_{it}\}$ a plan for the households $H = \{K_t, I_t, C_t, N_t\}$, a set of shocks governing investment technology $\{A_t\}$ subject to given prices and the set of shocks, F solves the firms' problem, and given prices, an initial amount of capital, and profits HP solves the household's problem, goods markets clear $Y_t = C_t + P_t I_t$, capital markets clear $k_t = K_t$ and labor markets clear $n_{yt} + n_{it} = N_t$

1.2 Part B

Derive the first-order conditions for the consumption good firm and the investment good firm. [5 pts]

For the consumption good firm:

$$K_t : r_t = \alpha(K_t/N_{yt})^{\alpha-1}$$

$$N_{yt} : w_t = (1-\alpha)(K_t/N_{yt})^{\alpha}$$

For the investment good firm:

$$N_{it} : P_t A_t = w_t$$

1.3 Part C

Set up the household Lagrangean and derive first-order conditions and the transversality condition. [5 pts]

$$\max_{C_t, K_{t+1}} \mathscr{L} = E_0 \sum_{t=0}^{\infty} \beta^t [1/2 C_t^{1/2} + \lambda_t [-C_t - P_t K_{t+1} + w_t + r_t K_t]]$$

$$C_t : 1/4 C_t^{-1/2} = \lambda_t$$

$$K_{t+1} : \lambda_t P_t = \beta E_t [\lambda_{t+1} r_{t+1}]$$

$$\lambda_t : BC$$

+ TVC $= \lim_{j \to \infty} E_t[\beta^{t+j} \lambda_{t+j} K_{t+j+1}] = 0$ "All dates all states"

1.4 Part D

Solve for the deterministic steady state. How does the steady state P depend on β? Provide intuition for this relationship. [12 pts]

Combine the equations from the HH problem to get the EE:

$$P = \beta r \tag{1}$$

From the Firms' problem we have:

$$r = \alpha(K/N_y)^{\alpha-1} \tag{2}$$

$$w = (1-\alpha)(K/N_y)^{\alpha} \tag{3}$$

$$w = PA \tag{4}$$

We have the BC:

$$C + PI = w + rK \tag{5}$$

And the law of motion

$$K = I \tag{6}$$

The goods market must clear

$$Y = K^{\alpha} N_y^{1-\alpha} = C \tag{7}$$

and

$$I = AN_i \tag{8}$$

And the labor market must clear

$$1 = N_i + N_y \tag{9}$$

Lastly in steady state there are no shocks so $A = 1$ (10). Our 10 unknowns are $C, P, I, K, Y, N_i, N_y, A, w, r$ so we can solve.

Combine the EE with the firms' problem:

$$(1-\alpha)(K/N_y)^{\alpha} = \beta\alpha(K/N_y)^{\alpha-1} \implies (K/N_y) = \frac{\beta\alpha}{1-\alpha}$$

This implies that P depends positively on β, a more patient economy drives the price of the investment good up since there is more demand for future consumption.

Combining 6+8+9 we get that

$$K = 1 - N_y$$

Substitute with the expression for K/N_y

$$\frac{1-N_y}{N_y} = \frac{\beta\alpha}{1-\alpha}$$

$$N_y = (\frac{\alpha\beta}{1-\alpha} + 1)^{-1}$$

From here, we can plug back in to find the rest of the variables we found along the way.

1.5 Part E

Use the log-linearized FOCs to solve for $\hat{N}_{yt}$ and $\hat{N}_{it}$ as a function of the state $\hat{K}_t$ and A_t and co-state $\hat{C}_t$. [8 pts]

Steps to log linearize:

Step 1 Take derivative of each side and evaluate at steady state

Step 2 Multiply by steady state variable

Step 3 Multiply by log-linearized variable

Step 4 Continue to next variable

Step 5 Simplify

Log linearize the firms' FOC's:

$$\hat{r}_t = (\alpha - 1)\hat{K}_t + (1 - \alpha)\hat{N}_{yt}$$

$$\hat{w}_t = \alpha\hat{K}_t - \alpha\hat{N}_{yt}$$

$$\hat{P}_t + \hat{A}_t = \hat{w}_t$$

And the BC

$$C\hat{C}_t + KP(\hat{P}_t + \hat{K}_{t+1}) = w\hat{w}_t + Kr(\hat{r}_t + \hat{K}_t)$$

Substitute these to find

$$\hat{N}_{yt} = \frac{C\hat{C}_t + \hat{K}_t(KP\alpha - w\alpha + Kr\alpha) - kP\hat{A}_t + kp\hat{K}_{t+1}}{KP\alpha - w\alpha + Kr(1 - \alpha)}$$

And we know that $N_{yt} = 1 - N_{it}$ so we have that

$$\hat{N}_{yt} = \frac{-N_i}{1 - N_i}\hat{N}_{it}$$

$$\hat{N}_{it} = \frac{N_i - 1}{N_i}\frac{C\hat{C}_t + \hat{K}_t(KP\alpha - w\alpha + Kr\alpha) - kP\hat{A}_t + kp\hat{K}_{t+1}}{KP\alpha - w\alpha + Kr(1 - \alpha)}$$

1.6 Part F

(15 points) Suppose we have log-linearized our model to attain

$$E_t x_{t+1} = Wx_t + G\hat{A}_t$$

where $x_t = \begin{bmatrix}\hat{K}_t \\ \hat{C}_t\end{bmatrix}$. Let $W = \begin{bmatrix}1 & -3 \\ 3 & -8.1\end{bmatrix}$, $J = \begin{bmatrix}-0.1 & 0 \\ 0 & -6.9\end{bmatrix}$ and $P^{-1} = \begin{bmatrix}1.3 & -0.5 \\ -0.5 & 1.3\end{bmatrix}$

Is there a locally unique solution? [2 pts]

Solve for $\hat{C}_t = \mu_{KC}\hat{K}_t + \mu_{AC}\hat{A}_t$ [8 pts]

Suppose $\hat{K}_0 = 0$ and $\hat{A}_0 = 1$. Solve for $\hat{C}_0$, $\hat{I}_0$ and explain intuitively. [5 pts]

- $n_u = n_y = 1$
- $R_{uy} = 1.3$ is invertible
- $|\lambda_i| \neq 1$

So yes there is a locally bounded solution.

To solve for bounded soln:

Step 1: Set up equation to solve for u_t

Step 2: Iterate forward one period

Step 3: Substitute u_{t+1}

Step 4: Find that the part with u_{t+j} drops to 0 since it is multiplied by a number less than 1 raised to an infinite exponent

Step 5: Factor out the exogenous part and substitute law of motion for exogenous shock

Step 6: Set up infinite sum and remember that $\sum_{t=0}^{\infty} \beta^t = \frac{1}{1-\beta}$

Step 7: Plug the result for u_t.

Remember our old friends:

$$x^*_{t+1} = Jx^*_t + P^{-1}G\hat{A}_t$$

and

$$x^*_t = P^{-1}x_t$$

So we get

$$\begin{bmatrix} s_{t+1} \\ u_{t+1} \end{bmatrix} = \begin{bmatrix} -0.1 & 0 \\ 0 & -6.9 \end{bmatrix} \begin{bmatrix} s_t \\ u_t \end{bmatrix} + \begin{bmatrix} 1.3 & -0.5 \\ -0.5 & 1.3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \hat{A}_t$$

$$u_{t+1} = -6.9u_t - 0.5\hat{A}_t$$

Step 1:

$$u_t = 1/6.9u_{t+1} + 0.5/6.9\hat{A}_t$$

Step 2:

$$E_t u_{t+1} = E_t[1/6.9u_{t+2} + 0.5/6.9\hat{A}_{t+1}]$$

Step 3:

$$u_t = 1/6.9(E_t[1/6.9u_{t+2} + 0.5/6.9\hat{A}_{t+1}]) + 0.5/6.9\hat{A}_t$$

Step 4:

$$u_t = 0.5/6.9 * \sum_{j=0}^{\infty} (1/6.9)^j E_t \hat{A}_{t+j}$$

We know that $\ln A_{t+1} = \rho \ln A_t + e_t$ so $E_t\hat{A}_{t+1} = \rho \ln A_t$

Step 5:

$$u_t = 0.5/6.9 * \hat{A}_t * \sum_{j=0}^{\infty} (1/6.9)^j * \rho^j$$

Step 6:

$$u_t = 0.5/6.9 * \hat{A}_t \frac{1}{1-\rho/6.9}$$

Step 7:

$$u_t = -0.5\hat{K}_t + 1.3\hat{C}_t$$

$$-0.5\hat{K}_t + 1.3\hat{C}_t = 0.5/6.9 * \hat{A}_t \frac{1}{1-\rho/6.9}$$

$$\hat{C}_t = \frac{1}{1.3}(0.5/6.9 * \hat{A}_t \frac{1}{1-\rho/6.9} + 0.5\hat{K}_t)$$

If $\hat{K}_0 = 0$ and $\hat{A}_0 = 1$

$$\hat{C}_0 = \frac{1}{1.3}(0.5/6.9 \frac{1}{1-\rho/6.9})$$

Then $\hat{I}_0 = \hat{K}_1$ so from the problem we have

$$\hat{I}_0 = \hat{K}_1 = 1 * \hat{K}_0 - 3\hat{C}_0 + \hat{A}_0$$

$$\hat{I}_0 = -3\frac{1}{1.3}(0.5/6.9 \frac{1}{1-\rho/6.9}) + 1$$

2 Final 22 Q1 Public Capital

Preferences are given by

$$\sum_{t=0}^{\infty} \beta^t \log c_t$$

where $c_t \geq 0$ is consumption. The HH has 1 unit of time that it supplies inelasitcally to the labor market. A firm produces the consumption good using technology $A_t k_t^{1/2} n_t^{1/2}$ where A_t is taken as given by the firm. Let w_t be the real wage and r_t be the rental rate of capital - both taken as given by the firm and household. A government provides a public capital in the amount P_t. This is a stock variable that evolves according to

$$P_{t+1} = G_t$$

where G_t is government purchases paid for with lump sum taxes $T_t = G_t$ collected from the household. G_t is endogenous and set to a fixed fraction of output $G_t = \gamma Y_t$ with $0 < \gamma < 1$. Note there is complete depreciation of

the public capital stock each period. The level of public capital determines A_t according to $A_t = \mu P_t^{1/2}$. The HH budget constraint is

$$c_t + I_t = w_t n_t + r_t k_t + \pi_t - T_t$$

The HH takes w_t, r_t, π_t, T_t as given with law of motion for private capital $K_{t+1} = I_t$.

2.1 Part A

Set up the HH problem and write down the Lagrangian, along with FOCs and TVC. [10 pts]

$$\max_{c_t, k_{t+1}} \mathscr{L} = \sum_{t=0}^{\infty} \beta^t [\log c_t + \lambda_t [w_t + r_t k_t + \pi_t - T_t - c_t - k_{t+1}]]$$

FOCs

$$c_t : 1/c_t = \lambda_t$$

$$k_{t+1} : \lambda_t = \beta \lambda_{t+1} r_{t+1}$$

Plus BC and TVC: $\lim \beta^t \lambda_t k_{t+1} = 0$

2.2 Part B

Write down the firm's problem along with the FOCs for profit max [5 pts]

$$\max_{k_t, n_t} \pi_t = A_t k_t^{1/2} r_t^{1/2} - r_t k_t - w_t n_t$$

$$k_t : 1/2 A_t (k_t/n_t)^{-1/2} = r_t$$

$$n_t : 1/2 A_t (k_t/n_t)^{1/2} = w_t$$

2.3 Part C

Solve for the balanced growth path competitive equilibrium along with labor's share (ratio of wage income to total income). [20 pts]

Under a balanced growth path the goal is to get the growth rates in terms of constants, and that the growth rates of output, consumption, capital should be the same, and wages should be constant (no growth). Also, the constraints on the parameters should be that $\gamma_c > 1$. So whenever you get the growth rate in terms of constants, plug them back into expression for growth rate and say it needs to be greater than 1.

1. Set up HH lagrangian
2. Take FOCs
3. Solve for EE (γ_c)

4. [1]Use other equations and definitions from problem to solve for growth rate. Then set the EE equal to the growth rate found from definitions in problem to solve for constant ratio of endogenous variables, or, if not applicable then you should get the growth rate in terms of parameters.

5. Use firm's production function to argue why $\gamma_y = \gamma_k$

6. Say that by the BC and TVC $\gamma_c = \gamma_y = \gamma_k$.

Step 3: Solving for the EE:

$$C_{t+1}/C_t = \beta r_{t+1}$$

$$\gamma_c = \beta 1/2 A_t k_t^{-1/2} = \beta 1/2 \mu P_t^{1/2} k_t^{-1/2}$$

This implies that along a balanced growth path the ratio of P_t/k_t is constant.
Step 4:

$$\gamma_P = \frac{P_{t+1}}{P_t} = \frac{G_t}{P_t} = \frac{\gamma Y_t}{P_t} = \frac{\gamma A_t k_t^{1/2}}{P_t} = \frac{\gamma \mu P_t^{1/2} k_t^{1/2}}{P_t}$$

$$\gamma_P = \gamma \mu (k/P)^{1/2}$$

Step 5:

$$\gamma_y = \frac{y_{t+1}}{y_t} = \frac{\mu P_{t+1}^{1/2} k_{t+1}^{1/2}}{\mu P_t^{1/2} k_t^{1/2}} = \gamma_P^{1/2} \gamma_k^{1/2}$$

So we get that $\gamma_y = \gamma_P = \gamma_k$
Step 6: By the BC and TVC we have $\gamma_c = \gamma_y = \gamma_P = \gamma_k$
Combining Steps 3 + 4:

$$\gamma \mu (k/P)^{1/2} = \beta 1/2 \mu (P/k)^{1/2}$$
$$P/k = \frac{2\gamma}{\beta}$$

Plugging in we get $\gamma_P = \gamma \mu (\frac{\beta}{2\gamma})^{1/2}$ To solve for labor share we take the ratio of wage income to total income:

$$\frac{w_t n_t}{y_t} = \frac{1/2 A_t k_t^{1/2}}{A_t k_t^{1/2}} = 1/2$$

2.4 Part D

Under what conditions, if any, is there economic growth?[10 pts]
Economic growth occurs when $\gamma_P > 1$ so when $\gamma \mu (\frac{\beta}{2\gamma})^{1/2} > 1$

[1]Note here that solving these problems sometimes is context specific. If you see something like $A_t = Y_{t-1}^{1/2}$ pretty good chance you should write out the definition of γ_A, and this should help you solve for the ratio of endogenous variables (think z/k or A/k).

2.5 Part E

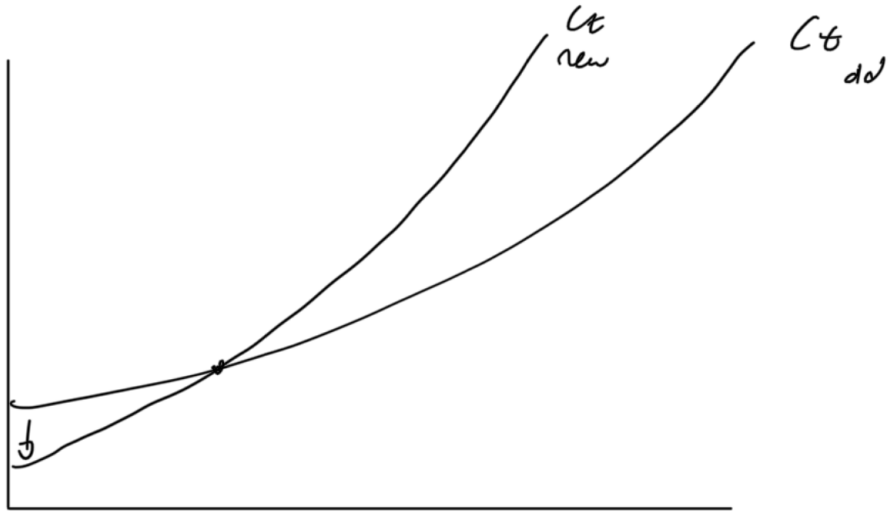
How does an increase in γ affect the level and growth rate of consumption? [10 pts]

γ increasing would increase the growth rate of consumption, and so the level of consumption would be higher as well.

2.6 Part F

Suppose this economy is on a balanced growth path. Would an increase in γ increase utility? [5 pts]

Initially it would decrease utility because $T_t = G_t = \gamma Y_t$ so a portion of income would be taken away. But the growth would be faster, so there is a breakeven point in the future where utility catches up to where it would have been under the old γ and surpasses it.



3 Aug 23 A Business Cycle Model with Gov Spending Shocks

The representative HH has preferences according to

$$E_0 \sum_{t=0}^{\infty} \beta^t [\ln C_t - N_t^2/2]$$

The household rents capital K_t to the firm at rate r_t and supplies labor N_t at the wage rate w_t. The household period t budget constraint is

$$C_t + I_t = w_t N_t + r_t K_t + \Pi_t - T_t$$

where C_t is consumption, I_t is investment, T_t is a lump-sum tax, and Π_t is any profit from the firm. The law of motion for capital is

$$K_{t+1} = (1-\delta)K_t + I_t$$

Gov spending is exogenous with

$$\hat{G}_{t+1} = \rho \hat{G}_t + e_{t+1}$$

where e_{t+1} is an iid mean zero process, $\hat{G}_t := \log G_t - \log \bar{G}$ and $\bar{G}$ is the steady state level of government spending set so steady state ratio of government spending to output is $\bar{G}/\bar{Y} = \mu$ with $0 < \mu < 1$. The government takes spending as given and sets the lump-sum tax rate $T_t = G_t$ each period. The firm seeks to maximize profit

$$\pi_t := K_t^{\alpha} N_t^{1-\alpha} - r_t K_t - w_t N_t$$

by choice of K_t, N_t taking r_t, w_t as given with $0 < \alpha < 1$.

3.1 Part A

Define a competitive equilibrium for this model. [5 pts]

A symmetric competitive equilibrium is a set of prices $P = \{w_t, r_t\}$, a plan for the firms $F = \{k_t, n_t, \pi_t\}$ a plan for the households $H = \{K_t, I_t, C_t, N_t\}$, and fiscal policy $FP = \{\hat{G}_t, T_t\}$ subject to given prices and FP, F solves the firms' problem, and given prices, an initial amount of capital, and profits HP solves the household's problem, goods markets clear $Y_t = C_t + I_t + G_t$, capital markets clear $k_t = K_t$ and labor markets clear $n_t = N_t$, and the government budget balances $G_t = T_t$.

3.2 Part B

Derive the first-order conditions for the firm. [5 pts]

$$K_t : \alpha K_t^{\alpha-1} N_t^{1-\alpha} = r_t$$

$$N_t : (1-\alpha) K_t^{\alpha} N_t^{-\alpha} = w_t$$

3.3 Part C

Set up the household Lagrangean and derive first-order conditions and the transversality condition. [5 pts]

$$\max_{C_t, K_{t+1}, N_t} \mathscr{L} = E_0 \sum_{t=0}^{\infty} \beta^t [\ln C_t - N_t^2/2 + \lambda_t [-C_t - K_{t+1} + w_t N_t + (r_t + 1 - \delta) K_t - T_t]]$$

$$C_t : 1/C_t = \lambda_t$$

$$K_{t+1} : \lambda_t = \beta E_t [\lambda_{t+1} (r_{t+1} + 1 - \delta)]$$

$$N_t : N_t = \lambda_t w_t$$

$$\lambda_t : BC$$

+ TVC = $\lim_{j \to \infty} E_t [\beta^{t+j} \lambda_{t+j} K_{t+j+1}] = 0$ "All dates all states"

3.4 Part D

Solve for the deterministic steady state. If μ increases, how does change the steady state K, N and C? Provide intuition for this relationship. [12 pts]

EE:

$$1 = \beta(r + 1 - \delta) \tag{10}$$

Consumption / Work trade off

$$CN = w \tag{11}$$

BC

$$C + I = wN + rK - T \tag{12}$$

LoM capital

$$\delta K = I \tag{13}$$

Steady state ratio of government expenditures:

$$G = \mu Y \tag{14}$$

FOCs from firm:

$$\alpha (K/N)^{\alpha - 1} = r \tag{15}$$

$$(1 - \alpha)(K/N)^{\alpha} = w \tag{16}$$

Government balances budget

$$G = T \tag{17}$$

Definition of output

$$Y = K^{\alpha} N^{1-\alpha} \tag{18}$$

Here our unknowns are $C, K, N, I, w, r, G, T, Y$ so there are 9 unknowns and 9 equations.

Solve to find

$$K/N = ((\beta^{-1} - 1 + \delta)\alpha^{-1})^{\frac{1}{\alpha - 1}}$$

And

$$N = \left(\frac{\alpha((\beta^{-1} - 1 + \delta)\alpha^{-1})^{\frac{\alpha}{\alpha-1}}}{((\beta^{-1} - 1 + \delta)\alpha^{-1})^{\frac{\alpha}{\alpha-1}}(\alpha - \mu) + ((\beta^{-1} - 1 + \delta)\alpha^{-1})^{\frac{1}{\alpha-1}}(\beta^{-1} - 1)}\right)^{1/2}$$